1.

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
 #include <sys/wait.h>
int main()
{
    char cmd[100];
    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if (pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if (pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
```

```c
    }
    return 0;
}
```

```
mahima@Mahima:~$ nano childprocess.c
mahima@Mahima:~$ gcc childprocess.c -o childprocess
mahima@Mahima:~$ ./childprocess
Enter Linux command: ls
Parent PID : 529
Child PID : 532
a.out           filename.c       forkexec    hello.txt           parentchildpid    username         wait.c
childprocess    filename.c.save  forkexec.c  parentchild         parentchildpid.c  username.c       waitt
childprocess.c  fork             hello       parentchild.c       process           username.c.save  waitt.c
filename        fork.c           hello.c     parentchild.c.save  process.c         wait
Child process completed.
```

```
mahima@Mahima:~$ ./childprocess
Enter Linux command: uname
Parent PID : 580
Child PID : 581
Linux
Child process completed.
```

```
mahima@Mahima:~$ ./childprocess
Enter Linux command: lscpu
Parent PID : 591
Child PID : 592
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   8
  On-line CPU(s) list:    0-7
Vendor ID:                GenuineIntel
  Model name:             11th Gen Intel(R) Core(TM) i5-1155G7 @ 2.50GHz
    CPU family:           6
    Model:                140
    Thread(s) per core:   2
    Core(s) per socket:   4
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             4992.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                          e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology
                           tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid s
                          se4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor la
                          hf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsb
                          ase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma
                           clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves vnmi avx5
                          12vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rd
                          pid movdiri movdir64b fsrm avx512_vp2intersect md_clear ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
```

```
  Hypervisor vendor:          Microsoft
  Virtualization type:        full
Caches (sum of all):
  L1d:                        192 KiB (4 instances)
  L1i:                        128 KiB (4 instances)
  L2:                         5 MiB (4 instances)
  L3:                         8 MiB (1 instance)
NUMA:
  NUMA node(s):               1
  NUMA node0 CPU(s):          0-7
Vulnerabilities:
  Gather data sampling:       Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:  Mitigation; Aligned branch/return thunks
  Itlb multihit:              Not affected
  L1tf:                       Not affected
  Mds:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Not affected
  Old microcode:              Not affected
  Reg file data sampling:     Not affected
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:       Not affected
  Spec store bypass:          Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:                 Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence
                              op, KVM SW loop
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
```

```
mahima@Mahima:~$ ./childprocess
Enter Linux command: lsblk
Parent PID : 604
Child PID : 605
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                               /
Child process completed.
mahima@Mahima:~$ ./childprocess
Enter Linux command: ps
Parent PID : 615
Child PID : 616
    PID TTY          TIME CMD
    316 pts/0    00:00:00 bash
    615 pts/0    00:00:00 childprocess
    616 pts/0    00:00:00 ps
Child process completed.
```

```
Enter Linux command: top
Parent PID : 628
Child PID : 629
top - 09:39:18 up 9 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   5799.6 total,   4704.9 free,    504.6 used,    733.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   5295.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24080  15412  11636 S   0.0   0.3   0:00.96 systemd
      2 root      20   0    3180   2212   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     46 root      19  -1   50400  16920  15636 S   0.0   0.3   0:00.21 systemd-journal
     78 systemd+  20   0   22540  14684  12064 S   0.0   0.2   0:00.10 systemd-resolve
     90 root      20   0   35292  12268   9248 S   0.0   0.2   0:00.28 systemd-udevd
    107 root      20   0    2888   2008   1884 S   0.0   0.0   0:00.06 chronyd-starter
    108 root      20   0    4472   3180   2932 S   0.0   0.1   0:00.01 cron
    109 message+  20   0    8820   5472   4804 S   0.0   0.1   0:00.07 dbus-daemon
    148 root      20   0   44092  29828  14688 S   0.0   0.5   0:00.41 networkd-dispat
    158 root      20   0   18436   9484   8236 S   0.0   0.2   0:00.12 systemd-logind
    163 root      20   0 1866288  15396  12612 S   0.0   0.3   0:00.12 wsl-pro-service
    166 syslog    20   0  220548   5904   4628 S   0.0   0.1   0:00.08 rsyslogd
    207 root      20   0    5492   2804   2620 S   0.0   0.0   0:00.00 agetty
    232 root      20   0  123456  32712  18104 S   0.0   0.6   0:00.21 unattended-upgr
    234 _chrony   20   0   24220  11880   7304 S   0.0   0.2   0:00.07 chronyd
    245 _chrony   20   0   12092   2496   1852 S   0.0   0.0   0:00.01 chronyd
    314 root      20   0    3184   1116    980 S   0.0   0.0   0:00.00 SessionLeader
    315 root      20   0    3200   1252   1108 S   0.0   0.0   0:00.08 Relay(316)
    316 mahima    20   0    6272   5596   3928 S   0.0   0.1   0:00.07 bash
```

2.

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if (src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while ((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
```

```c
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
mahima@Mahima:~$ nano file_copy.c
mahima@Mahima:~$ gcc file_copy.c -o file_copy
mahima@Mahima:~$ nano input.txt
mahima@Mahima:~$ ./file_copy
File copied successfully.
mahima@Mahima:~$ cat output.txt
Hello World
This is OSSP Lab.
```